

Technical Document #214: Software Engineering - Comprehensive Overview

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Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Domain-driven design (DDD) models software around business domains.
- Code review examines code changes before merging.
- Test-driven development (TDD) writes tests before writing code.